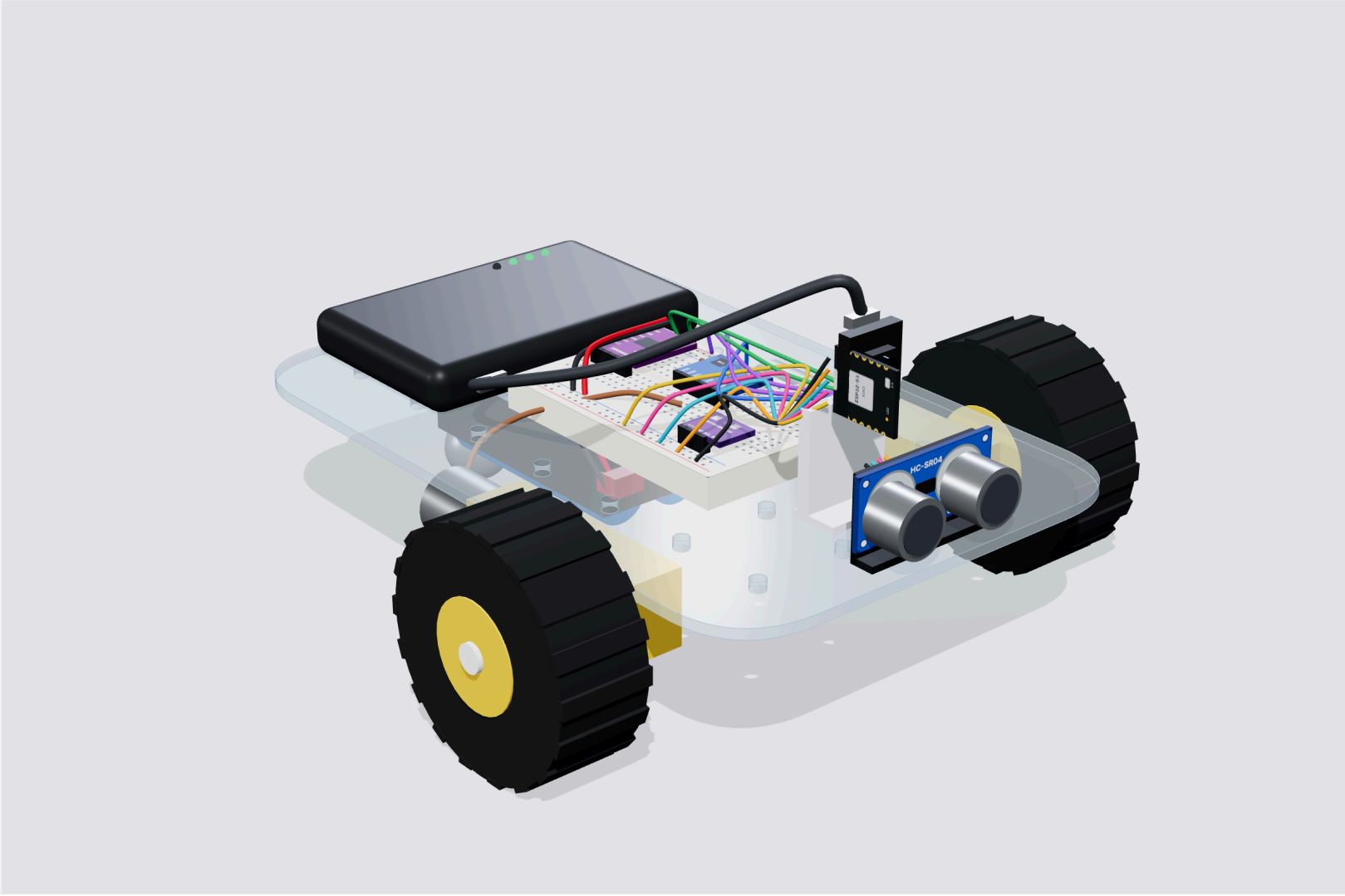
RAQIB · THE FINISHED ROBOT · 3D VIEW

The finished robot

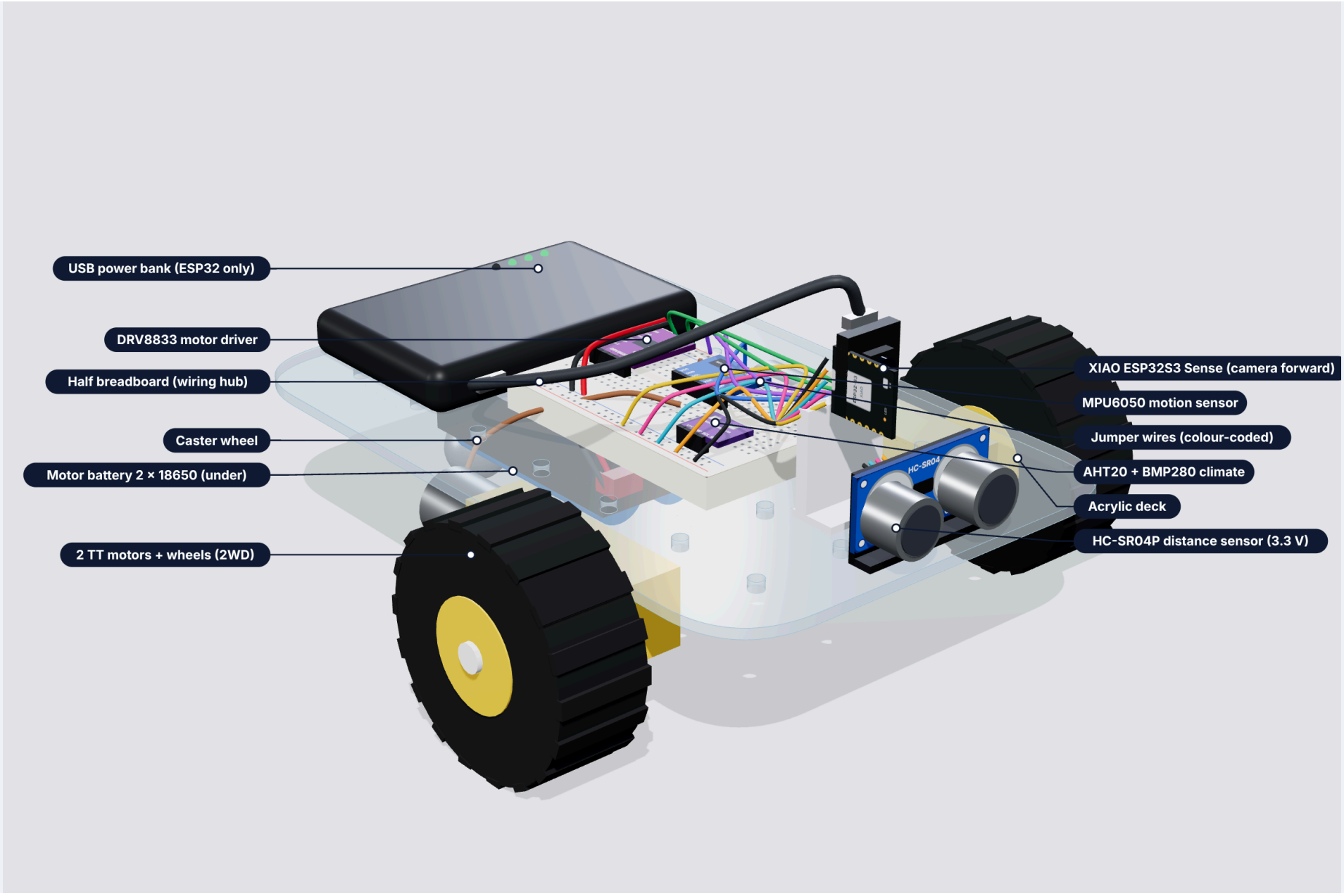
Raqib 3.0, the easy ESP32 inspection robot, fully built: one acrylic deck, two-wheel drive with a caster, a XIAO ESP32S3 Sense (camera + microphone) standing at the front over one distance sensor, and every other part on a half breadboard. No soldering of the sensors, no Raspberry Pi: your laptop is the brain.



A 3D model at real size (mm), built from the parts and layout of the build guide. Board colours and printed labels follow the common versions of each part; your boards may differ slightly.

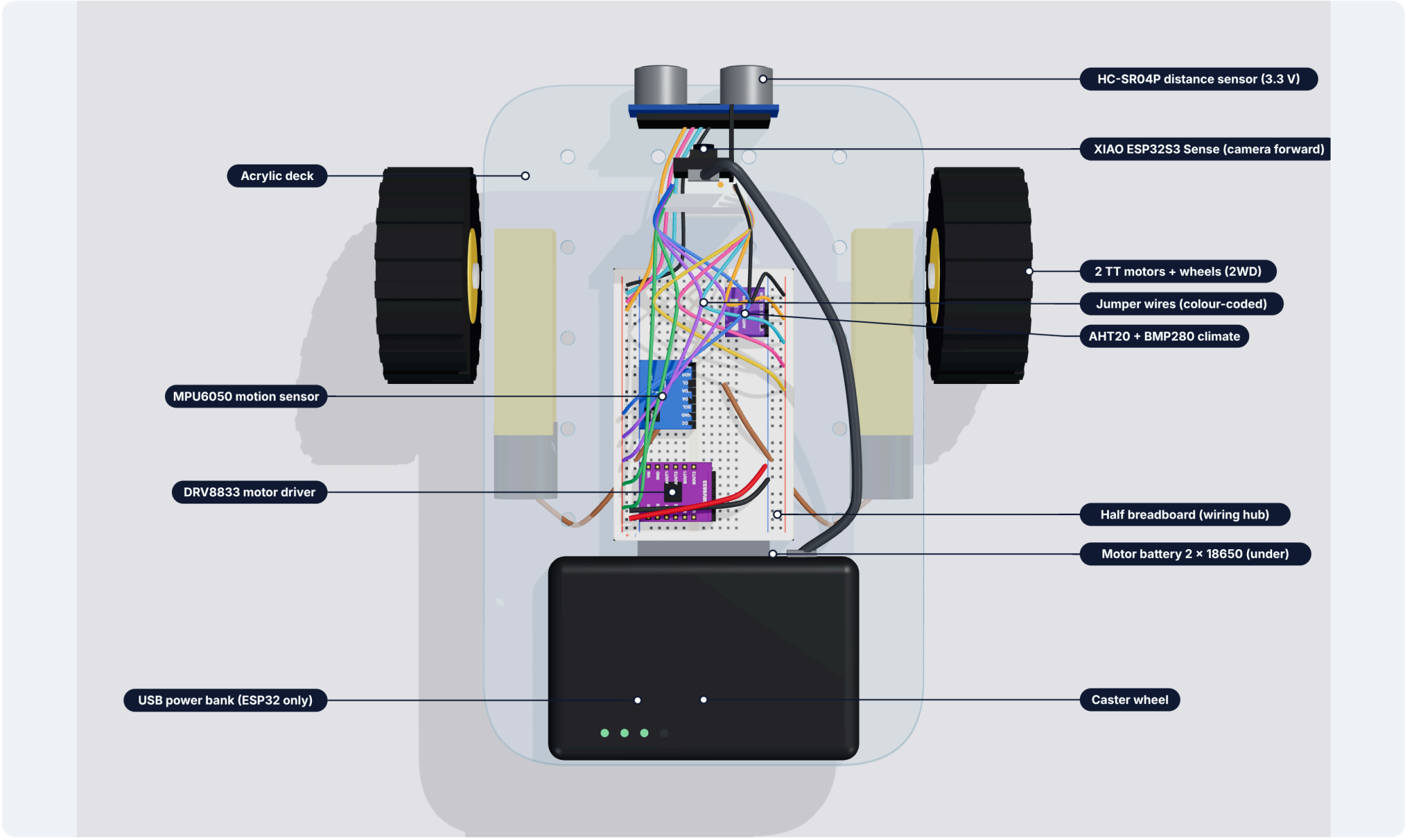
Every part, labelled

All parts in their final place. Front of the robot = the side with the camera and the distance sensor.



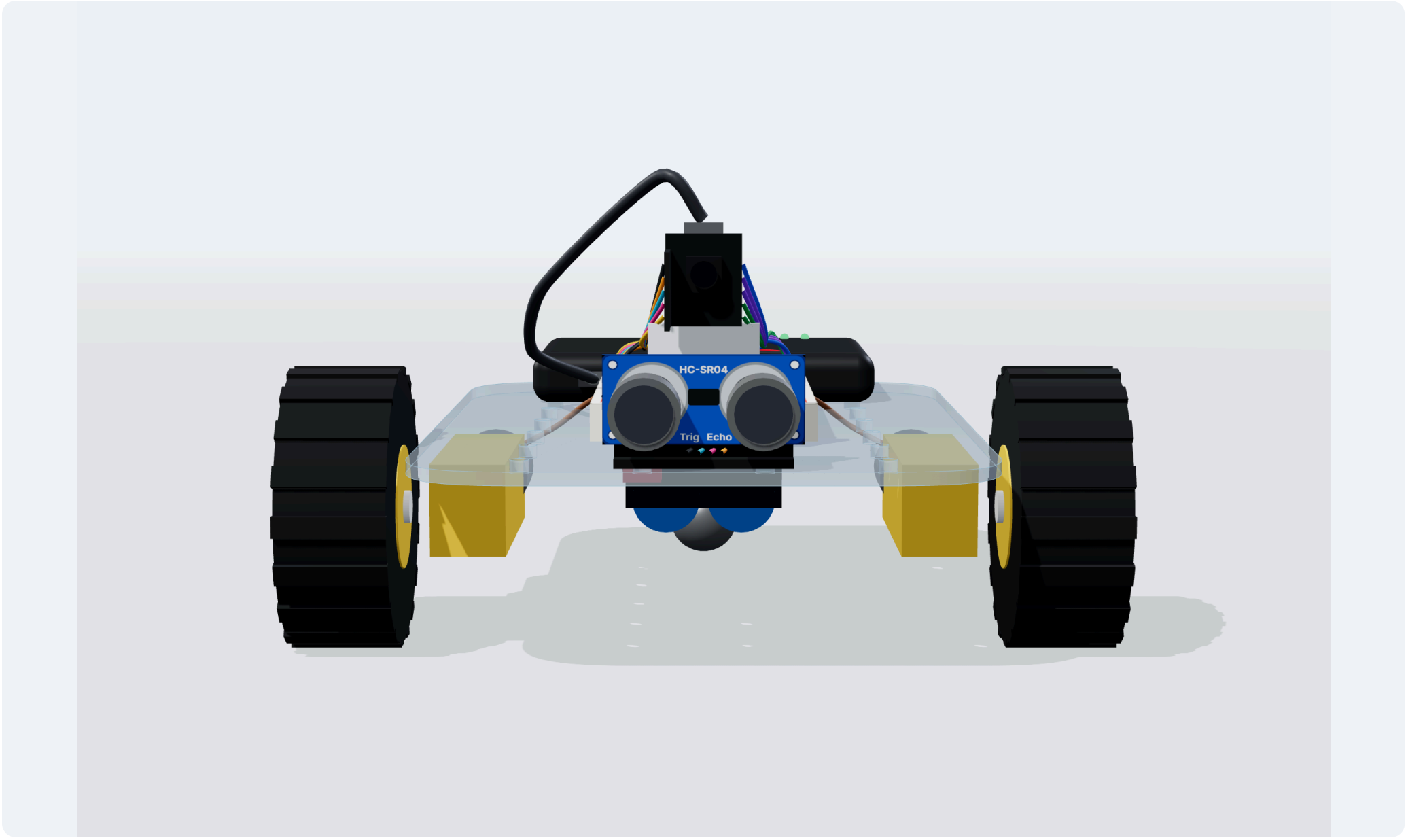
Seen from above: how the deck is organised

Front: the HC-SR04P at the edge and the XIAO on its bracket just behind it, camera looking forward. Middle: the half breadboard with the DRV8833, MPU6050 and AHT20 + BMP280 plugged in. Back: the USB power bank, which powers only the XIAO.



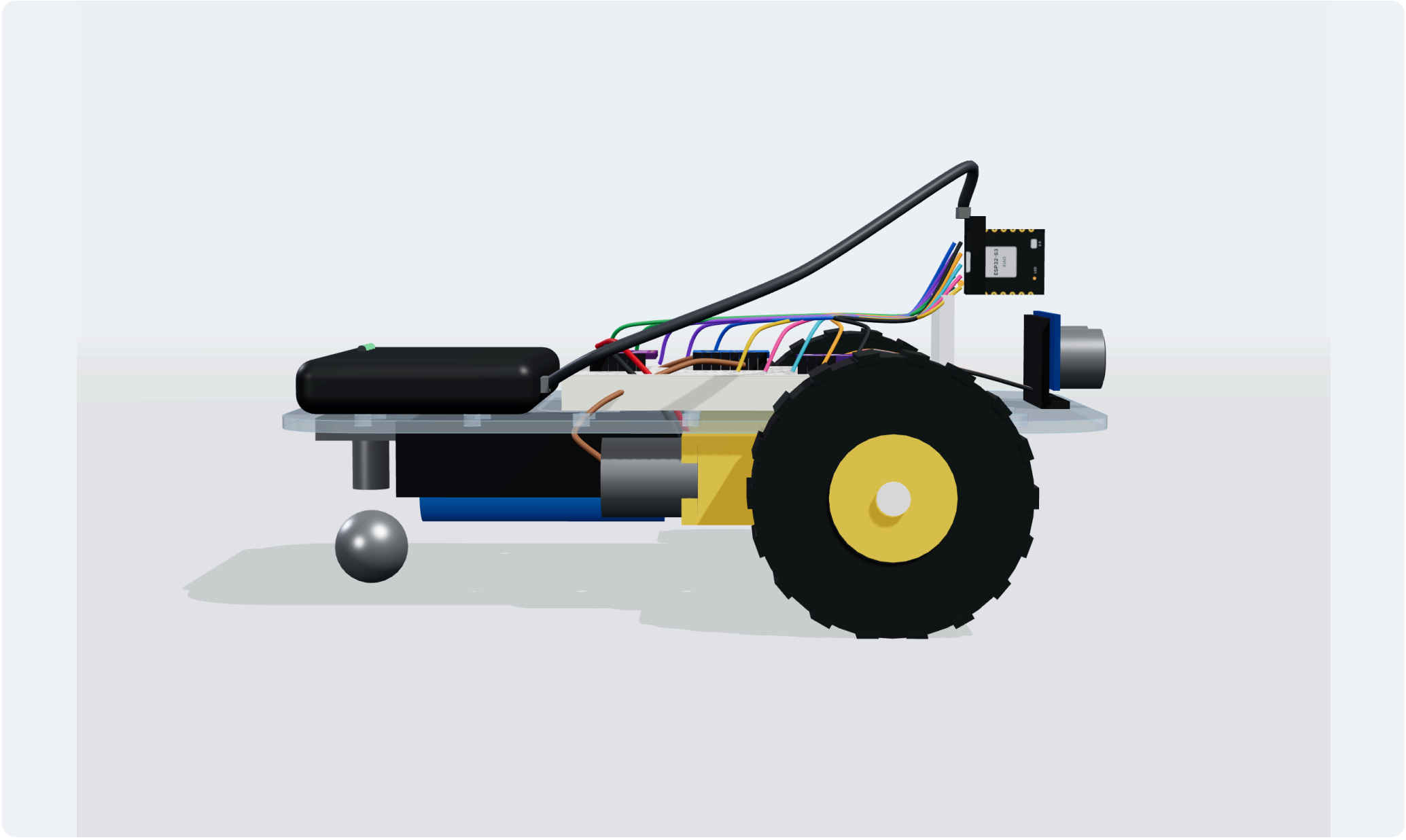
Front view

What the robot looks at: the distance sensor straight ahead and the camera just above it. The camera is fixed: the robot turns its whole body to look around.



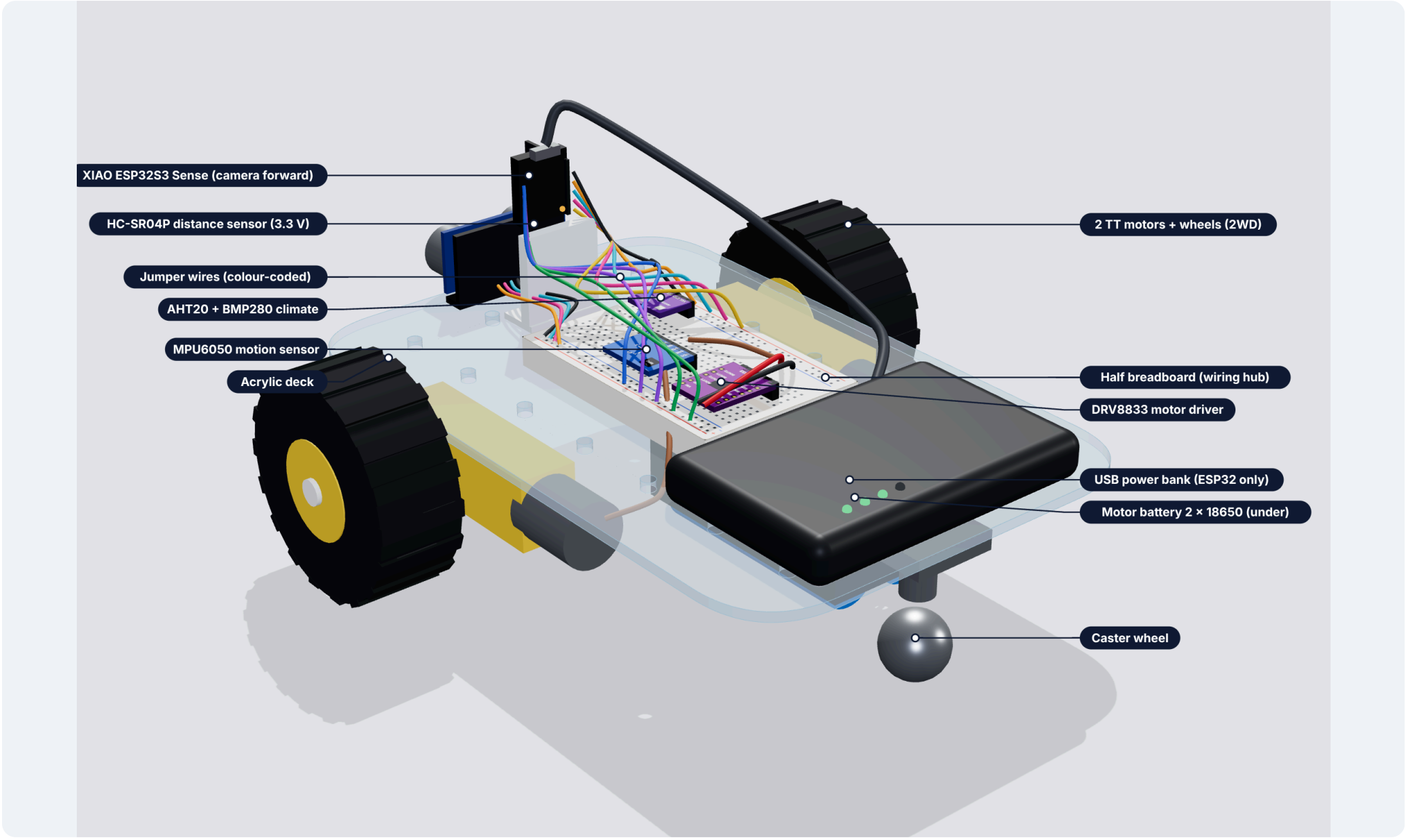
Side view

One deck. On top: power bank, breadboard, XIAO and sensor. Under it: the 2 × 18650 motor battery and the two TT motors at the front; the caster ball at the back.



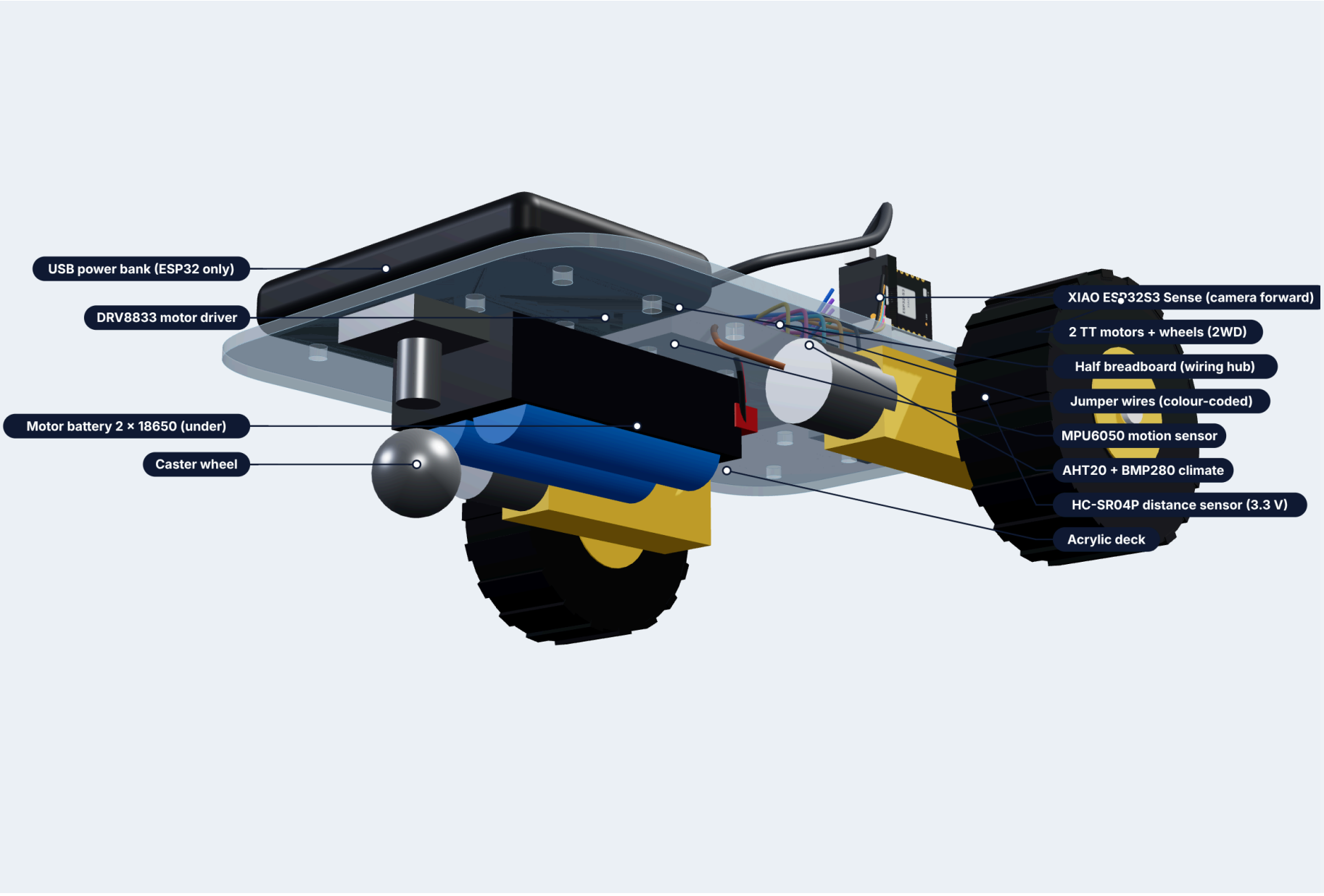
From the back

The power bank's USB-C cable goes to the top of the XIAO. The motor battery's red and black leads come up through the deck to the DRV8833 (VM and GND); the DRV8833's GND row is also joined to the 3V3 / GND rails, so everything shares one ground.



Underneath

The 2 × 18650 holder sits under the deck with its switch reachable and powers only the motors. Two TT motors at the front (left and right), the caster at the back. No encoders: the robot uses its gyro to turn and times its motors.



Close-up of the wiring

Colour-coded jumper wires from the XIAO down to the breadboard: green / purple motor direction (D0–D3), blue SDA (D4), yellow SCL (D5), pink TRIG (D6), cyan ECHO (D7), orange 3V3, black GND.

